TRAUMA · NEET-SS / FET / INI-SS

Tibial plateau and pilon

1. Tibial plateau fractures

Anatomy and mechanism

- The **lateral plateau** is **convex** and higher; the **medial plateau** is **concave**, larger and stronger (more load).
- Mechanism: **valgus load with axial force** (lateral plateau, commonest), varus (medial plateau, higher energy, often **fracture-dislocation**).
- **Lateral plateau fractures are commonest (about 55-70%).**
- Associated injuries: **lateral meniscus tears** (commonest with lateral plateau; often peripheral and incarcerated), **medial meniscus**, **MCL and ACL injuries**, **peroneal nerve** (lateral fracture-dislocation), **popliteal artery** (medial plateau and Schatzker IV/VI, bicondylar fractures; **ankle-brachial index under 0.9 = arterial study**), **compartment syndrome** (especially Schatzker VI).

Schatzker classification

Type	Description	Typical patient / mechanism	Treatment principle
I	Lateral **split** (wedge)	Young, dense bone	Percutaneous screws or buttress plate; check for meniscal entrapment
II	Lateral **split-depression**	**Commonest**	Elevate the depressed fragment, **subchondral bone graft/substitute**, raft screws, lateral buttress plate

Type	Description	Typical patient / mechanism	Treatment principle
III	Lateral **pure depression**	**Osteoporotic** elderly	Elevation through a cortical window, graft, raft screws (IIIa lateral, IIIb central)
IV	**Medial plateau**	High energy; fracture-dislocation (Moore); **popliteal artery and peroneal nerve at risk**	Posteromedial approach, buttress plate; poor prognosis
V	**Bicondylar** with metaphysis intact	High energy	Dual plating / lateral locking ± medial plate
VI	Bicondylar with **metaphyseal-diaphyseal dissociation**	Highest energy; **compartment syndrome**	**Staged**: spanning external fixator, then ORIF (dual incisions) when soft tissues allow; or hybrid/circular frame

- **Three-column concept (Luo; CT-based):** lateral, medial and **posterior** columns (the posterior column, especially **posteromedial fragments**, is missed by Schatzker; needs a **posteromedial (Lobenhoffer) or posterior approach** and buttress plating).
- **Moore classification** (fracture-dislocations): split, entire condyle, rim avulsion, rim compression, four-part. Fracture-dislocations carry more ligament and meniscal injuries and **neurovascular injury rising from about 2% (type I) to about 50% (type V), overall about 15%** (similar to a knee dislocation).

Investigation

- AP, lateral, oblique; **CT** for every operative plateau; **MRI** shows meniscal/ligament injuries (commonly present).

- **Assess soft tissues** (fracture blisters, swelling: **"wrinkle sign"** shows readiness for surgery, usually 7-14 days).

Management

- **Non-operative:** undisplaced or minimally displaced, stable to varus/valgus stress in full extension (under about 10 degrees), elderly low-demand.
- **Operative indications: articular step-off over 2-3 mm** (debated: stability and alignment matter more), **condylar widening over 5 mm**, **instability (over 10 degrees varus/valgus in extension)**, **medial plateau fractures** (inherently unstable), open fractures, compartment syndrome, vascular injury.
- **Principles:** restore the **mechanical axis and joint stability** first (more important than perfect articular congruity), elevate the joint surface, **support it with subchondral rafting screws and bone graft/substitute** (calcium phosphate cement resists subsidence best), **buttress** the metaphysis, repair menisci.
- **Approaches:** anterolateral (lateral plateau; submeniscal arthrotomy to see the joint); posteromedial (Lobenhoffer, between the medial gastrocnemius and pes anserinus/semimembranosus); posterior (Burks-Schaffer, Carlson); avoid a single midline incision with extensive stripping for bicondylar fractures (**dual incisions with a skin bridge over about 7 cm** historically; soft tissue is the priority).
- **Staged protocol for high-energy fractures: spanning external fixation** (ligamentotaxis), CT in the frame, definitive fixation once the soft tissues settle.
- **Arthroscopy-assisted reduction (ARIF)** for lateral split/depression fractures.
- **Elderly with severe comminution/arthritis:** acute TKA in selected cases.

Complications

- **Infection and wound breakdown** (bicondylar, early surgery through poor soft tissue), **post-traumatic arthritis** (malalignment and instability more than step-off), **stiffness/arthrofibrosis**, loss of reduction (subsidence), **compartment syndrome**, **peroneal nerve injury**, DVT, malunion.

2. Tibial shaft fractures (brief link)

- Commonest long-bone fracture; **highest rate of open fractures** and compartment syndrome.
- **Reamed intramedullary nail** for most displaced fractures (**SPRINT**: no difference between reamed and unreamed overall; possible benefit of reaming in closed fractures). **Suprapatellar (semi-extended) nailing** eases proximal and distal third fractures (less anterior knee pain debated).
- **Proximal third fractures** tend to **valgus and procurvatum (apex anterior)** with nailing: use **blocking (Poller) screws**, a more lateral/proximal start point, extended positioning, plates.
- **Distal third**: nail with blocking screws, or plate; fix the fibula if needed for alignment.
- Non-operative (Sarmiento functional bracing) for low-energy, stable fractures with acceptable alignment (under 5 degrees varus/valgus, under 10 degrees AP angulation, under 1 cm shortening, under 10 degrees rotation).
- **Anterior knee pain** is the commonest complication after tibial nailing.

3. Tibial pilon (plafond) fractures

Mechanism and features

- **Axial loading**: the talus is driven into the distal tibial articular surface (falls from height, road traffic accidents). Distinct from rotational ankle fractures.

- **Soft-tissue injury is the dominant problem** (thin anterior and medial soft tissue envelope).
- Fragments (Chaput's): typically **medial**, **anterolateral (Chaput, attached to AITFL)**, **posterolateral (Volkmann, attached to PITFL)**, with central die-punch impaction; associated fibula fracture in about 75%.
- **Associated injuries:** spine fractures, calcaneus fractures, contralateral injuries (falls from height).

Classification

Ruedi-Allgower	Description
I	Undisplaced cleavage fracture of the joint
II	Displaced articular fracture, minimal comminution
III	Comminuted and impacted articular surface

- **AO/OTA 43:** A extra-articular, B partial articular, C complete articular (C3 = comminuted articular + metaphyseal).
- **Tscherne** for closed soft-tissue injury (C0-C3).

Management: the staged protocol

1. **Stage 1 (urgent): spanning external fixator** (tibia to calcaneus/first metatarsal) to restore length, alignment and rotation by ligamentotaxis; ± **fibular fixation** (if the plafond team and soft tissue plan allow; malreduced fibula makes later tibial reduction impossible).
2. **CT in the frame**: plan approaches.
3. **Stage 2 (when soft tissues allow, usually 7-21 days; "wrinkle sign", blisters re-epithelialised): ORIF** of the articular surface with plates (anterolateral, anteromedial, posterolateral, medial approaches depending on fragments), bone graft for metaphyseal defects.

4. **Rüedi-Allgöwer principles:** (1) restore fibular length, (2) reconstruct the articular surface, (3) bone graft the metaphyseal defect, (4) buttress the medial tibia.
5. **Alternatives:** definitive **circular (Ilizarov) or hybrid external fixation** with limited internal fixation when soft tissues are poor; **primary ankle arthrodesis** for unreconstructable fractures.
6. **Approaches: anterolateral** (between the EDL/peroneus tertius and the fibula; superficial peroneal nerve at risk), **anteromedial** (medial to tibialis anterior; thinnest soft tissue), **posterolateral** (between peroneals and FHL; sural nerve), **medial**. Skin bridges of at least 5-7 cm between incisions (debated).

Complications

- **Wound dehiscence and deep infection** (historically up to 30-40% with early ORIF; staged protocols reduced this to under 5-10%).
- **Post-traumatic ankle arthritis** (common; related to the initial cartilage damage more than reduction quality), malunion (varus), non-union (metaphyseal), stiffness, chronic pain, complex regional pain syndrome.

> Exam pearl: **Schatzker II (split-depression) is the commonest; IV (medial) = fracture-dislocation with popliteal artery and peroneal nerve risk; VI = metaphyseal-diaphyseal dissociation with compartment syndrome.** The **lateral meniscus** is the commonest associated soft-tissue injury.

> Exam pearl: **Pilon = axial load; soft tissue dictates timing; stage it: span, scan, plan, then fix (7-21 days).** Rüedi-Allgöwer's four steps start with **fibular length**.

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.